

Amjad Althabteh

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SUMMARY

Systems-level C++ engineer who builds things close to the metal. Authored a custom OpenGL rendering engine, orbital simulation system, and C++23 diagnostics toolchain from scratch. Drawn to problems where memory layout, GPU throughput, and pipeline architecture actually matter.

EDUCATION

Wayne State University
B.S. in Computer Science

Detroit, MI
GPA: 3.7

EXPERIENCE

Immersalab Graphics | Detroit, MI

09/25 – 02/26

Graphics Engineer

- Optimized GPU resource handling and rendering pipeline stability.
- Designed internal C++/JSON tooling and asset pipeline; profiled GLSL shader compilation paths.
- Reduced GPU idle time by ~30% through rendering pipeline optimizations.

UE Game Developer | Ann Arbor, MI

04/25 – 09/25

Unreal Engine C++ Developer

- Architected a gameplay ability system in C++ handling 10+ character actions with Unreal's GAS framework
- Built animation state machines for 3+ characters managing 15+ blend states with smooth transition logic
- Optimized animation systems enabling smooth multi-layer blending across 15+ states.

Rare Munchies | Ann Arbor, MI

09/23 – 04/24

Software Development Intern

- Optimized client-side rendering pipeline, reducing unnecessary re-renders by ~35% through state architecture improvements.
- Built reusable TypeScript component systems focused on runtime efficiency and minimal overhead.

Wayne State University | Detroit, MI

02/25 – 12/25

Undergraduate Researcher — Orbital Collision Modeling

- Modeled 50+ satellite trajectories using numerical integration and N-body gravitational dynamics.
- Accelerated large-scale orbital simulations using spatial partitioning, reducing computing time by ~40% across 1M+ trajectory evaluations.

PROJECTS

Glint 3D Rendering Engine | C++17/OpenGL

[glint-rendering](#)

- Designed a fixed-size memory pool allocator using placement new and free list reuse to minimize heap allocation overhead.
- Optimized rendering performance through frustum culling and GPU instanced rendering, reducing unnecessary draw calls across scenes.

3D Space Simulation Engine

[orbitalcollision.dev](#)

- Engineered a modular C++17/OpenGL simulation engine for orbital systems.
- Developed a custom 3D math library supporting vector and matrix transformations for rendering and simulation systems.

HFT OrderBook Engine | C++20 | Performance Systems

- Built a C++20 order matching engine processing high-throughput bid/ask workflows with **microsecond-level latency**.
- Eliminated false sharing and reduced memory contention using cache-aligned (alignas(64)) structures.

Traceforge — Diagnostics Engine | Stack Trace & Memory Tracking

- Engineered a C++23 diagnostics engine using std::stacktrace to improve runtime debugging visibility.
- Implemented custom allocation tracking using overloaded new/delete operators for memory leak detection.

SKILLS

Languages: C++ (C++11–C++23), Rust, C, C#, Python, TypeScript, Java, SQL

Graphics & Engines: OpenGL, GLSL, Unreal Engine, Unity, Godot

Systems: Custom allocators, memory pools, multithreading, lock-free data structures, cache optimization, profiling